ROLL NO: | | | | | | | | | | |

(To fill the roll no. is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE-AI)

Semester: VI

SUBJECT CODE: BCSAI612

BACKEND DEVELOPMENT

Time: 3 Hours **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION- A

NOTE: Attempt all parts of the following questions. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is the purpose of the scripts section in a package.json file?

(ii) What is the data type of the _id field in MongoDB by default?

(iii) Which method is used to retrieve all documents from a MongoDB collection using Mongoose?

(iv) Mention the difference between Schema and Model in Mongoose.

(v) What is the purpose of require() in Node.js?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is the purpose of the $Match stage in MongoDB aggregation?

(ii) Write a short note on updateOne() and updateMany() methods in MongoDB.

(iii) Which method in Express is used to register middleware?

(iv) How do you define a schema in Mongoose?

(v) Write the Mongoose method to insert multiple documents at once.

SECTION-B

NOTE: Attempt all the questions. (5 × 2 × 2 = 20 Marks)

Q.3.(i)(a) Write the code to use Multer in your project.

OR

(b) What is SetTimeout() in node.js? Explain with an example.

(ii)(a) Write the code to use Promise.all.

OR

(b) Write a MongoDB aggregation pipeline to calculate the total number of users registered per city from a user's collection with fields name, email, and city.

Q.4.(i)(a) Which node.js package is used for jwt authentication? Write the code.

OR

(b) Create a NodeJS server using express with 5000 port.

(ii)(a) What is the use of Promise.allSettle? Write the code.

OR

(b) Write any 5 methods of objects in javascript.

SECTION-C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Differentiate in detail between SQL & No-SQL.

OR

(b) Explain the working of req.params, req.query, and req.body in Express with examples.

Q.6.(a) Explain the error handling in node.js.

OR

(b) What is the Multer package in NodeJS? Explain in brief.